OligoTox-Hydrocephalus

An open, provenance-resolved dataset of hydrocephalus and CSF-dynamics toxicity for oligonucleotide therapeutics

NIH/NCATS Oligonucleotide Toxicity Open Data Challenge — Phase 2 (Data Generation) · Narrative document · release v1.0

Measurements	Oligonucleotides	Sources	Trials	Position records	QC checks
1,361	53	193	155	256	53/53 pass

1 Executive summary

OligoTox-Hydrocephalus pairs the design of 53 oligonucleotide therapeutics with 1,361 measured hydrocephalus and CSF-dynamics outcomes drawn from 193 sources. It is the first public dataset for this endpoint. Hydrocephalus is the eighth toxicity on the Challenge's list of interest and, on our reading of the Phase 1 cohort, no winning team addressed it.

The endpoint is unusually hostile to naive curation, and the dataset's design is a direct response to three hazards. **The disease causes the endpoint** — the populations dosed with CNS oligonucleotides have elevated baseline rates of exactly this outcome. **The delivery procedure causes the endpoint** — repeated lumbar puncture produces CSF leak and low-pressure states in placebo arms as much as treated arms. **Almost nobody images the ventricles**, so for most compounds the absence of a finding means nobody looked. Each hazard is answered by a column that a modeller can filter on, not by a caveat in a document.

Positive and negative controls

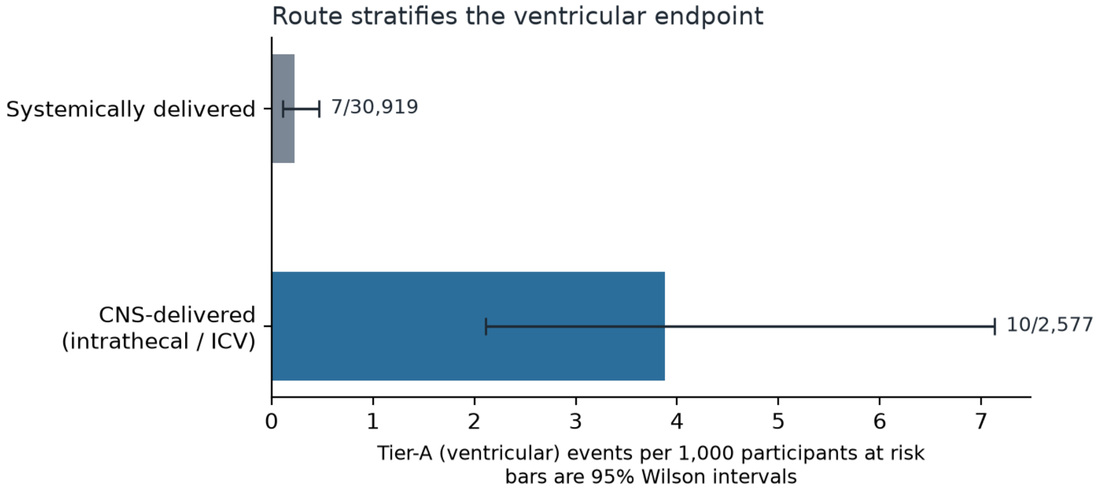
The dataset is not a list of toxic compounds. Its negative class is larger and more deliberate than its positive class, which is what makes it trainable.

- **755 tier-A explicit measured negatives.** Not absences: arms where the endpoint term is listed with a reported count of zero, or where 42 CFR 11.48(a)(4)(ii)(A) requires the serious-adverse-event table to be complete and no ventricular term appears in it.
- **64 trials with a concurrent placebo or sham arm**, contributing 8,071 comparator participants — the strongest design available here.
- **A route-contrast class.** Systemically dosed oligonucleotides are carried deliberately so the delivery hypothesis is testable rather than assumed.
- **Disease background rates carrying no compound at all** (`tox_axis = disease_background_rate`).
- **Procedure-attributable events** (`delivery_procedure_complication`), which occur in placebo arms and must be excluded from any compound-level analysis.
- **A designed negative control** — a scrambled non-targeting siRNA — and **two protective-direction rows**, an siRNA that *prevents* ventriculomegaly, kept on their own axis and left ungraded so a benefit can never be read as an absent harm.

2 Main findings and conclusions

Finding 1 — delivery route stratifies the endpoint, strongly, between cohorts

Across 546 trial arms and 36,324 participants at risk, the tier-A (ventricular) event rate is **3.88 per 1,000** (95% CI 2.11–7.13) for CNS-delivered oligonucleotides against **0.23 per 1,000** (0.11–0.47) for systemically delivered ones — odds ratio 17.2, Fisher exact $p = 8.4\text{e-}08$.

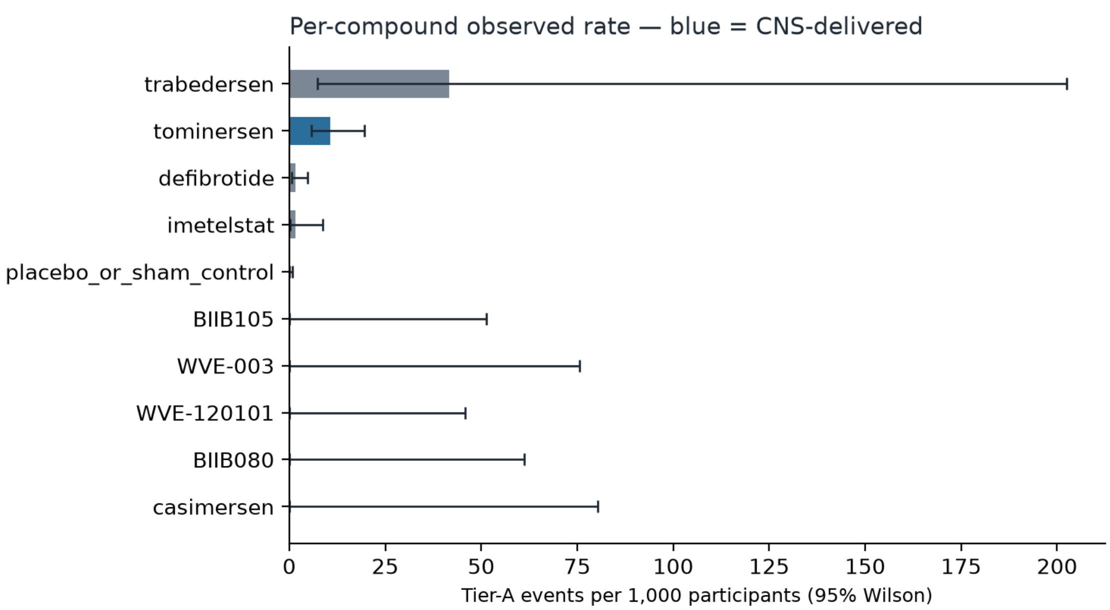


Finding 2 — and that contrast disappears inside randomised comparisons

This is the finding we would most want a reader to carry away. Restricting to the 64 trials that contain their own concurrent comparator arm, treated arms report 2 tier-A events in 15,633 participants against 2 in 8,071 comparator participants — odds ratio 0.52, p = 0.61. **No detectable within-trial effect.** The large between-route contrast in Finding 1 is a comparison between different diseases, different ages and different monitoring intensities, not between treated and untreated people. Both numbers are in the dataset, and reporting only the first would be the single easiest way to mislead with it.

Finding 3 — the ventricular signal concentrates in one compound, and it was measured

Serious hydrocephalus or normal-pressure hydrocephalus appears in three separate tominersen studies, including 2/263 against 0/264 in the concurrent placebo arm of GENERATION HD1. The phase 1/2a trial made ventricular volume a pre-specified MRI outcome: from screening to day 197 the placebo arm moved 35.58 → 36.46 mL (+2.5%, n=12) while the two highest dose arms moved +13.0% (n=9) and +19.9% (n=10); in the open-label extension the ventricular-volume boundary shift integral rose 46.1% on monthly against 18.8% on bimonthly dosing. These are the only rows in the dataset where the ventricles were *measured* rather than incidentally observed.



Finding 4 — but the endpoint is not compound-specific, and the two attributed cases differ mechanistically

In an n-of-1 protocol, both infants dosed intrathecally with valeriasen — a different ASO, target, indication and age group — developed ventricular enlargement; one required endoscopic third ventriculostomy at a CSF opening pressure of 55 cmH_2O, the other an external drain and then a shunt. The authors call it "a potential monitorable toxicity of some intrathecal antisense oligonucleotides". The mechanisms diverge: the tominersen index case was attributed to a sterile meningitis with CSF protein 2.64 g/L and lymphocytosis, whereas the KCNT1 patients had a negative CSF inflammatory panel and a dose-related hypothesis — evidenced by a reduced-dose rechallenge, delivered without recurrence. The dataset keeps them apart on `tox_axis` and `event_cluster_id` rather than pooling them into one count.

Finding 5 — the disease is a fourfold confounder

In the era before nusinersen was approved, SMA patients had a hydrocephalus incidence rate ratio of 4.7 (95% CI 2.4–10.2) against matched non-SMA controls. Those rows are in the dataset, carrying no compound. Any analysis of an SMA-indicated oligonucleotide that omits them will over-attribute.

3 How the data were produced

No wet-lab experiment was performed; this is a curation, and the methods are of two kinds kept strictly separate — the experimental methods of the source studies, and the curation methods used here. The full account is the companion Methodology document. In outline:

Modality	Route used	What it contributes
Trial registry	ClinicalTrials.gov v2 API	Per-arm event counts **with denominators and comparator arms**, including explicitly reported zeros. Trial selection is a query over every oligonucleotide with posted results — 155 trials — not a hand-picked list.
Pharmacovigilance	openFDA FAERS	Post-marketing reporting proportions across the marketed class.
Regulatory labelling	DailyMed SPL; EMA SmPC	Verbatim, regulator-adjudicated risk statements with their section codes.
Primary literature	Europe PMC fullTextXML	The two drug-attributed index cases and their mechanisms.
Disease epidemiology	Europe PMC fullTextXML	Untreated-population incidence — the confounder control.
Chemistry	WHO INN Recommended lists	Sequences and per-position chemistry, by deterministic parse.

Computational processing. Six of the eight build components are deterministic parsers over payloads committed to the repository, so their values cannot be mistyped and can be re-derived offline. The largest defect class found in a review of our sibling kidney dataset was hand-transcription error; a parser cannot make it, and its mapping tables are auditable in one place. Every network call is cached, so the whole dataset rebuilds from a clean checkout with one command sequence.

4 Indicators, predictor variables and their distribution

The response variable. `hydroceph_grade` is an ordinal 0–3 severity with a rubric written for this endpoint, not inherited from another organ. Every graded row carries in `grade_basis` the exact rule that produced it. The scale is censored by study type — a CSF-composition row cannot reach grade 3, which is defined by whole-organism intervention — so grades are not comparable across `study_type`, and we say so rather than letting it be discovered.

Grade	0	1	2	3	not graded

Rows	1,133	101	79	22	26

Ascertainment is recorded, not assumed. The column separates *measured_positive*, *measured_null* (actively assessed and not found, 1,133 rows), *reported_threshold_limited* and *not_assessed*. The QC suite **enforces** that a grade of 0 occurs only where ascertainment is measured_null. A model that ignores this column will learn the reporting process rather than the biology.

Attribution is the source's, never ours. `attribution_as_stated` is *not_discussed* for 1,325 rows because registry and pharmacovigilance records carry no causality assessment at all. That is a property of those sources. Attribution for them must be read structurally, from the comparator arm and the disease baseline — which is why both are in the dataset.

Predictor variables across the tested oligonucleotides

Design predictors are carried at chemistry and design level for all 53 compounds and at **per-position** level for 13: 256 rows giving the sugar, base, 5-methylation and phosphorothioate-versus-phosphodiester status at every nucleotide. 13 compounds carry a published sequence. The remainder — the double-stranded siRNAs, the morpholinos and the compounds that enter only through the trial registry — do not, and that is the dataset's principal limitation for sequence-level modelling.

Subject class	Rows
animal_in_vitro	2
animal_in_vivo	8
human_in_vivo	1,348
human_population	3
in vitro (any species)	**0**

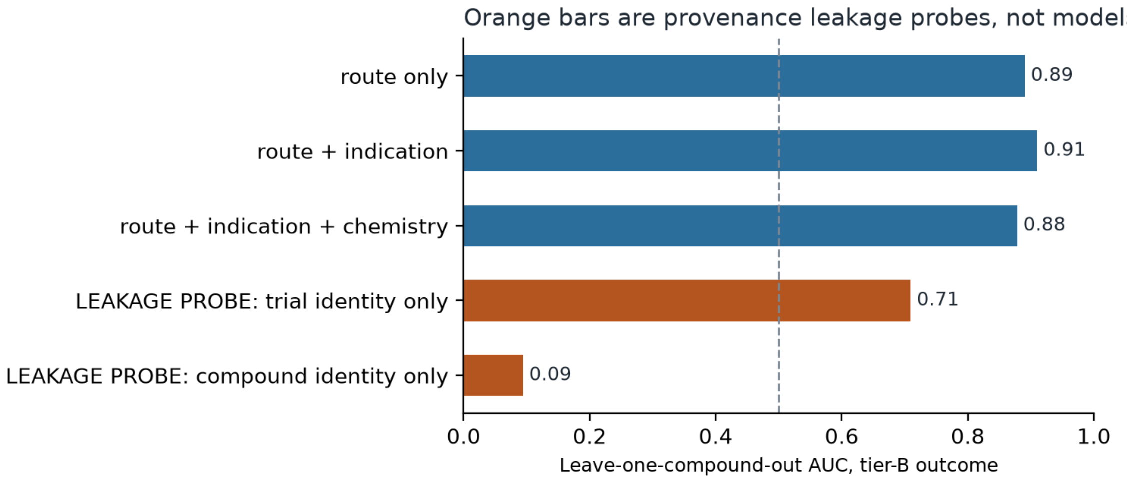
Human and animal evidence are separated by a single column, `subject_class`, which the QC suite re-derives and fails on any disagreement, with generated human-only and animal-only views. **The dataset contains no in vitro rows.** The Challenge states that datasets based on in vitro human systems, or able to extrapolate between in vitro human systems and animal data, are of particular interest; this release does neither, and we would rather state that plainly than imply otherwise.

5 The gap this addresses

Before this release there was no public, structured dataset pairing oligonucleotide identity with hydrocephalus or CSF-dynamics outcomes. The evidence existed but was scattered across registry adverse-event tables, spontaneous-report databases, two regulators' labels, and a handful of case reports — in formats that cannot be joined and with no shared severity scale. The specific gap this closes is not that the events were unknown; it is that **nobody could see them next to their own denominators, their comparator arms and their disease baselines at the same time**. That juxtaposition is what turns a set of alarming case reports into something a model can be fitted to, and it is what produced Finding 2 — a null within-trial contrast that the case literature alone would never have surfaced.

6 Using this data to build a predictive model

We fitted models rather than merely proposing them, and report what they do and do not support. The unit of analysis is the **trial arm**, not the measurement row: rows within an arm are not independent, and an arm carries a denominator. Validation is **leave-one-compound-out**, because arms of the same compound are correlated and a random split leaks the compound across folds.



On the tier-B (CSF-dynamics) outcome, which occurs in 84 arms and is the mechanistic precursor the index case documents, route and indication give a leave-one-compound-out AUC of 0.91 (bootstrap 95% CI 0.848–0.951). Adding chemistry *degrades* it, because chemistry is NOT_REPORTED for most compounds and contributes noise. The tier-A outcome, with 9 affected arms, will not support a classifier and we do not present one.

The leakage probes are the most useful result. A model given only trial identity reaches AUC 0.709, so a material share of any apparent performance is provenance, not biology. A model given only compound identity scores 0.094 — below chance, which is the correct behaviour under leave-one-compound-out and confirms the validation is doing its job. We ran these because a review of our sibling kidney dataset found study_type and source_id were strong shortcut predictors of its label.

What a user should build with it, and what they should not

- **Supported:** route- and population-stratified risk models; separating drug effect from procedure and disease effect; modelling ascertainment explicitly as a covariate; hypothesis generation for CSF-dynamics monitoring in intrathecal programmes.
- **Not supported:** sequence-to-toxicity prediction across the full roster (13 of 53 sequences); within-compound dose–response for tier A; in vitro-to-in vivo extrapolation; and any causal claim about an individual compound, given Finding 2.

The most valuable next experiment this dataset points to is not a bigger model. It is a human in vitro choroid-plexus or blood-CSF-barrier system in which a PS-oligonucleotide is applied and CSF-relevant transport is measured — because the one axis that would let anybody extrapolate from these clinical observations to a new compound is precisely the axis on which no public data exists.

Availability. Dataset, build and QC code, and every retrieved source payload: `github.com/naviero1/Oligos, toxicity/hydrocephalus/`. Curation released under CC BY 4.0; per-row rights in the redistribution column (1,322 rows public domain). All 53 QC checks pass on the released revision.